

Exercise 2.4

Let

$$S_0(x) = \exp \left\{ - \left(Ax + \frac{1}{2} Bx^2 + \frac{C}{\log D} D^x - \frac{C}{\log D} \right) \right\},$$

where A , B , C , and D are all positive.

- (a) Show that the function S_0 is a survival function.
- (b) Derive a formula for $S_x(t)$.
- (c) Derive a formula for μ_x .
- (d) Now suppose that

$$A = 0.00005, \quad B = 0.0000005, \quad C = 0.0003, \quad D = 1.07.$$

- (i) Calculate ${}_t p_{30}$ for $t = 1, 5, 10, 20, 50, 90$.
- (ii) Calculate ${}_t q_{40}$ for $t = 1, 10, 20$.
- (iii) Calculate ${}_{t|10} q_{30}$ for $t = 1, 10, 20$.
- (iv) Calculate e_x° for $x = 70, 71, 72, 73, 74, 75$.
- (v) Calculate e_x° for $x = 70, 71, 72, 73, 74, 75$, using numerical integration.

Solution

(a)

To show that S_0 is a survival function, we verify the main criteria:

$$S_0(0) = 1, \quad S_0(x) \text{ is nonincreasing}, \quad \lim_{x \rightarrow \infty} S_0(x) = 0.$$

First,

$$\begin{aligned} S_0(0) &= \exp \left\{ - \left(A(0) + \frac{1}{2} B(0)^2 + \frac{C}{\log D} D^0 - \frac{C}{\log D} \right) \right\} \\ &= \exp \left\{ - \left(\frac{C}{\log D} - \frac{C}{\log D} \right) \right\} \\ &= \exp\{0\} \\ &= 1. \end{aligned}$$

Now define

$$H(x) = Ax + \frac{1}{2} Bx^2 + \frac{C}{\log D} D^x - \frac{C}{\log D},$$

so that

$$S_0(x) = e^{-H(x)}.$$

Then

$$H'(x) = A + Bx + CD^x.$$

Since $A > 0$, $B > 0$, $C > 0$, and $D^x > 0$, we have

$$H'(x) > 0 \quad \text{for } x \geq 0.$$

Therefore, $H(x)$ is increasing, and hence

$$S'_0(x) = -H'(x)e^{-H(x)} < 0.$$

Thus $S_0(x)$ is decreasing on $x \geq 0$.

Finally, for the usual mortality case $D > 1$,

$$\lim_{x \rightarrow \infty} H(x) = \infty,$$

because the terms Ax , $\frac{1}{2}Bx^2$, and $\frac{C}{\log D}D^x$ grow without bound. Hence,

$$\lim_{x \rightarrow \infty} S_0(x) = \lim_{x \rightarrow \infty} e^{-H(x)} = 0.$$

Therefore, S_0 satisfies the criteria for a survival function.

(b)

For a life aged x , the survival function for an additional t years is

$$S_x(t) = {}_t p_x = \frac{S_0(x+t)}{S_0(x)}.$$

Using the given expression for S_0 , we obtain

$$S_x(t) = \frac{\exp \left\{ - \left(A(x+t) + \frac{1}{2}B(x+t)^2 + \frac{C}{\log D}D^{x+t} - \frac{C}{\log D} \right) \right\}}{\exp \left\{ - \left(Ax + \frac{1}{2}Bx^2 + \frac{C}{\log D}D^x - \frac{C}{\log D} \right) \right\}}.$$

Combining the exponents,

$$\begin{aligned} S_x(t) &= \exp \left\{ - \left[A(x+t) + \frac{1}{2}B(x+t)^2 + \frac{C}{\log D}D^{x+t} - \frac{C}{\log D} \right] + \left[Ax + \frac{1}{2}Bx^2 + \frac{C}{\log D}D^x - \frac{C}{\log D} \right] \right\} \\ &= \exp \left\{ -At - \frac{1}{2}B((x+t)^2 - x^2) - \frac{C}{\log D}(D^{x+t} - D^x) \right\}. \end{aligned}$$

Therefore,

$$S_x(t) = \exp \left\{ -At - \frac{1}{2}B((x+t)^2 - x^2) - \frac{C}{\log D}(D^{x+t} - D^x) \right\}.$$

(c)

The force of mortality is

$$\mu_x = -\frac{1}{S_0(x)} \frac{d}{dx} S_0(x).$$

From part (a), $S_0(x) = e^{-H(x)}$, where

$$H(x) = Ax + \frac{1}{2}Bx^2 + \frac{C}{\log D}D^x - \frac{C}{\log D}.$$

Then

$$H'(x) = A + Bx + CD^x.$$

Therefore,

$$\begin{aligned} \frac{d}{dx} S_0(x) &= \frac{d}{dx} e^{-H(x)} \\ &= -H'(x) e^{-H(x)} \\ &= -S_0(x) (A + Bx + CD^x). \end{aligned}$$

Substituting into the definition of μ_x gives

$$\begin{aligned} \mu_x &= -\frac{1}{S_0(x)} [-S_0(x) (A + Bx + CD^x)] \\ &= A + Bx + CD^x. \end{aligned}$$

Thus,

$$\boxed{\mu_x = A + Bx + CD^x}.$$

(d)(i)

Using the formula from part (b), with

$$A = 0.00005, \quad B = 0.0000005, \quad C = 0.0003, \quad D = 1.07,$$

we have, for a life aged 30,

$${}_t p_{30} = S_{30}(t) = \exp \left\{ -At - \frac{1}{2}B((30+t)^2 - 30^2) - \frac{C}{\log D}(D^{30+t} - D^{30}) \right\}.$$

Thus,

$$\begin{aligned}
{}_1p_{30} &= \exp \left\{ -A - \frac{1}{2}B(31^2 - 30^2) - \frac{C}{\log D}(D^{31} - D^{30}) \right\} \\
&= 0.997574991, \\
{}_5p_{30} &= \exp \left\{ -5A - \frac{1}{2}B(35^2 - 30^2) - \frac{C}{\log D}(D^{35} - D^{30}) \right\} \\
&= 0.986177871, \\
{}_{10}p_{30} &= \exp \left\{ -10A - \frac{1}{2}B(40^2 - 30^2) - \frac{C}{\log D}(D^{40} - D^{30}) \right\} \\
&= 0.967229794, \\
{}_{20}p_{30} &= \exp \left\{ -20A - \frac{1}{2}B(50^2 - 30^2) - \frac{C}{\log D}(D^{50} - D^{30}) \right\} \\
&= 0.906413014, \\
{}_{50}p_{30} &= \exp \left\{ -50A - \frac{1}{2}B(80^2 - 30^2) - \frac{C}{\log D}(D^{80} - D^{30}) \right\} \\
&= 0.381218527, \\
{}_{90}p_{30} &= \exp \left\{ -90A - \frac{1}{2}B(120^2 - 30^2) - \frac{C}{\log D}(D^{120} - D^{30}) \right\} \\
&= 0.000000351.
\end{aligned}$$

Therefore,

t	1	5	10	20	50	90
${}_tp_{30}$	0.997574991	0.986177871	0.967229794	0.906413014	0.381218527	0.000000351

(d)(ii)

For a life aged 40,

$${}_tq_{40} = 1 - {}_tp_{40}.$$

Using the formula from part (b),

$${}_tp_{40} = S_{40}(t) = \exp \left\{ -At - \frac{1}{2}B((40+t)^2 - 40^2) - \frac{C}{\log D}(D^{40+t} - D^{40}) \right\}.$$

Therefore,

$$\begin{aligned}
{}_1q_{40} &= 1 - \exp \left\{ -A - \frac{1}{2}B(41^2 - 40^2) - \frac{C}{\log D}(D^{41} - D^{40}) \right\} \\
&= 0.004706934, \\
{}_{10}q_{40} &= 1 - \exp \left\{ -10A - \frac{1}{2}B(50^2 - 40^2) - \frac{C}{\log D}(D^{50} - D^{40}) \right\} \\
&= 0.062877281, \\
{}_{20}q_{40} &= 1 - \exp \left\{ -20A - \frac{1}{2}B(60^2 - 40^2) - \frac{C}{\log D}(D^{60} - D^{40}) \right\} \\
&= 0.174725068.
\end{aligned}$$

Thus,

t	1	10	20
${}_tq_{40}$	0.004706934	0.062877281	0.174725068

(d)(iii)

The deferred probability of death is

$${}_t|_{10}q_{30} = {}_tp_{30} {}_{10}q_{30+t}.$$

Using the same parameter values, for $t = 1$ we have

$$\begin{aligned}
{}_1|_{10}q_{30} &= {}_1p_{30} {}_{10}q_{31} \\
&= 0.997574991 \left(1 - \exp \left\{ -10A - \frac{1}{2}B(41^2 - 31^2) - \frac{C}{\log D}(D^{41} - D^{31}) \right\} \right) \\
&= 0.997574991(0.034982718) \\
&= 0.034897884.
\end{aligned}$$

For $t = 10$,

$$\begin{aligned}
{}_{10}|_{10}q_{30} &= {}_{10}p_{30} {}_{10}q_{40} \\
&= 0.967229794 \left(1 - \exp \left\{ -10A - \frac{1}{2}B(50^2 - 40^2) - \frac{C}{\log D}(D^{50} - D^{40}) \right\} \right) \\
&= 0.967229794(0.062877281) \\
&= 0.060816780.
\end{aligned}$$

For $t = 20$,

$$\begin{aligned}
{}_{20}|_{10}q_{30} &= {}_{20}p_{30} {}_{10}q_{50} \\
&= 0.906413014 \left(1 - \exp \left\{ -10A - \frac{1}{2}B(60^2 - 50^2) - \frac{C}{\log D}(D^{60} - D^{50}) \right\} \right) \\
&= 0.906413014(0.119352337) \\
&= 0.108182511.
\end{aligned}$$

Thus,

t	1	10	20
$t _{10}q_{30}$	0.034897884	0.060816780	0.108182511

(d)(iv)

The curtate expectation of life is

$$e_x = \sum_{k=1}^{\infty} {}_k p_x.$$

Using the formula from part (b),

$${}_k p_x = \exp \left\{ -Ak - \frac{1}{2}B((x+k)^2 - x^2) - \frac{C}{\log D}(D^{x+k} - D^x) \right\}.$$

Therefore,

$$e_x = \sum_{k=1}^{\infty} \exp \left\{ -Ak - \frac{1}{2}B((x+k)^2 - x^2) - \frac{C}{\log D}(D^{x+k} - D^x) \right\}.$$

Evaluating this sum for $x = 70, 71, 72, 73, 74, 75$ gives

x	70	71	72	73	74	75
e_x	13.046404	12.517404	12.001471	11.498671	11.009049	10.532638

(The process using the table is in the Excel file)

(d)(v)

The complete expectation of life is

$$e_x^{\circ} = \int_0^{\infty} {}_t p_x dt.$$

Using numerical integration, this may be approximated from the curtate expectation of life by

$$e_x^{\circ} \approx e_x + \frac{1}{2}.$$

Using the values from part (d)(iv), we obtain

$$\begin{aligned} e_{70}^{\circ} &\approx 13.046404 + 0.5 = 13.546404, \\ e_{71}^{\circ} &\approx 12.517404 + 0.5 = 13.017404, \\ e_{72}^{\circ} &\approx 12.001471 + 0.5 = 12.501471, \\ e_{73}^{\circ} &\approx 11.498671 + 0.5 = 11.998671, \\ e_{74}^{\circ} &\approx 11.009049 + 0.5 = 11.509049, \\ e_{75}^{\circ} &\approx 10.532638 + 0.5 = 11.032638. \end{aligned}$$

Therefore,

x	70	71	72	73	74	75
e_x°	13.546404	13.017404	12.501471	11.998671	11.509049	11.032638